Beyond Power Laws: What Scaling Reveals About Operator Learning

Lucas Perrier

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Abstract

Empirical scaling laws have guided resource allocation in large-scale machine learning, but their applicability to operator learning—where discretization resolution introduces a third scaling axis alongside dataset size and model capacity—remains poorly understood. We conduct a comprehensive study across three PDE benchmarks (Burgers, Darcy, diffusion), three architectures (MLP, DeepONet, FNO), and 5,040 training runs spanning two orders of magnitude in each axis. Rather than assuming power-law relationships *a priori*, we fit six candidate functional forms (power, exponential, logarithmic, stretched exponential, saturation, linear) per scaling axis and use the corrected Akaike Information Criterion (AICc) to let the data select the governing law.

Our key findings challenge several common assumptions: (i) *data scaling follows logarithmic rather than power-law decay* in 7 of 9 task–model combinations, implying slower-than-algebraic diminishing returns from additional training data; (ii) *resolution scaling is task-dependent and non-universal*—higher resolution helps for advection-dominated problems (Burgers: 35/35 slices improve), is mixed for elliptic problems (Darcy), and actively *increases error* for smooth problems (Diffusion: error rises in 27/35 slices for MLP and DeepONet); (iii) *capacity scaling is largely negligible* within our range, with most configurations already in the capacity-sufficient regime; and (iv) the irreducible error floor E_∞ extracted from the winning fits provides an architecture-dependent error floor estimate that ranks models independently of data volume or resolution.

These results demonstrate that the *type* of scaling law, not just its exponents, provides additional diagnostic information about the interaction between PDE structure, model architecture, and discretization in this setting.

1 Introduction

Empirical scaling laws have become a central tool for understanding and predicting the performance of large-scale machine learning systems [**?** **?**]. In natural language processing and computer vision, power-law relationships between test loss and dataset size, parameter count, and compute budget guide resource allocation decisions worth millions of dollars.

Scientific machine learning (SciML) faces an analogous resource allocation problem with a twist: the practitioner must additionally choose the *discretization resolution* at which training data are generated. Solving PDEs at higher resolution is expensive—often the dominant cost—yet the relationship between resolution and downstream operator-learning error is poorly characterized.

Neural operators [**?** **?**] learn mappings between infinite-dimensional function spaces and are often marketed as "resolution-invariant." Two prominent architectures—DeepONet [**?**] and the Fourier Neural Operator (FNO) [**?**]—use fundamentally different strategies to handle varying discretizations. Despite these architectural innovations, it remains unclear *when* resolution invariance

translates into practical benefits. Does doubling the training resolution halve the error? Does it matter more than doubling the dataset? How does the answer depend on the PDE?

A subtler question, largely absent from the literature, is whether power laws are even the *right functional form* for operator-learning scaling. The power-law assumption $E \propto X^{-\alpha}$ originates from statistical learning theory and information-theoretic arguments in specific regimes. When the error–resource relationship follows a different functional form—logarithmic, exponential, saturating—forcing a power-law fit obscures the true scaling behavior and leads to misleading exponent estimates.

Contributions.

1. We introduce **multi-law model selection for scaling analysis**: rather than assuming power laws, we fit six candidate functional forms (power, exponential, logarithmic, stretched exponential, saturation, linear) and use AICc to identify the governing law per axis, per task, per model. The *identity* of the winning law becomes a physics-interpretable diagnostic.

2. We report **5,040 training runs** and **13,440 cross-resolution evaluations** across three PDE families, three architectures, and seven capacity levels.

3. We find that **data scaling is predominantly logarithmic** ($E = a - b \ln N$) rather than algebraic for 7 of 9 task–model combinations, with only Diffusion FNO/MLP exhibiting genuine power-law data scaling ($\alpha \approx 1.1$).

4. We show that **resolution scaling is non-universal**: it helps consistently only for Burgers, is mixed for Darcy, and *actively hurts* for Diffusion (MLP, DeepONet). This challenges the assumption that resolution is a uniformly beneficial scaling axis.

5. We demonstrate that **estimated irreducible error floors** E_∞ extracted from the best per-axis fits provide architecture-dependent error floor estimates: FNO achieves $E_\infty = 0.003$ on Diffusion vs. DeepONet's 0.19—a 63× gap that persists regardless of data volume.

2 Related Work

Neural scaling laws. **?**] and **?**] established power-law scaling of cross-entropy loss with dataset tokens, parameter count, and compute for autoregressive language models. **?**] showed similar trends across vision and speech. In scientific ML, **?**] introduced scaling-law diagnostics for physics-informed approaches (PINNs, HNNs) in low-dimensional settings, finding that physics priors shift the error floor rather than the scaling exponent. Our work extends this program to infinite-dimensional operator learning, elevates resolution to a scaling variable, and—critically—relaxes the power-law assumption via multi-law model selection.

Methodological positioning. The dominant methodology in neural scaling laws assumes a power-law functional form *a priori* and focuses on estimating exponents and proportionality constants [**?** **?** **?**]. Our work adopts a complementary perspective: rather than assuming power laws, we treat the *identity* of the best-fitting functional form as itself informative. This family-selection-centric view does not discard the traditional approach; instead, it includes power laws as one of six candidates and explicitly tests when the traditional baseline succeeds (Diffusion FNO/MLP) and when it fails (Burgers / Darcy data scaling, where logarithmic fits dominate).

Operator learning. DeepONet [?] builds on the universal approximation theorem for operators [?], factorizing the learned operator into branch and trunk components. The Fourier Neural Operator (FNO) [?] parameterizes integral kernels via truncated Fourier series, achieving state-of-the-art accuracy on several PDE benchmarks. Recent extensions include U-shaped architectures [?], attention mechanisms [?], and physics-informed variants [?].

Resolution and discretization in operator learning. A key selling point of neural operators is their ability to generalize across resolutions. ?] demonstrated FNO's zero-shot super-resolution capability. However, systematic studies of how *training* resolution affects accuracy—and whether more resolution is always better—are lacking. Our work fills this gap, and our finding that higher resolution can *increase* error challenges the prevailing wisdom.

3 Multi-Law Scaling Framework

3.1 Problem Setting

We consider the supervised learning of solution operators $\mathcal{G}^\dagger : \mathcal{U} \to \mathcal{V}$, where $\mathcal{U}$ and $\mathcal{V}$ are Banach spaces of functions on some domain $\Omega \subset \mathbb{R}^d$. Given N input–output pairs $\{(u_i, v_i)\}_{i=1}^N$ with $v_i = \mathcal{G}^\dagger(u_i)$, a model $\mathcal{G}_\theta$ with $D = |\theta|$ parameters is trained on data discretized at spatial resolution R (grid points per spatial dimension).

3.2 Beyond Power Laws: Candidate Functional Forms

Prior work on neural scaling laws has assumed the power-law form $E(X) = E_\infty + a\,X^{-\alpha}$, motivated by learning-theoretic bounds. We relax this assumption and consider six candidate laws for describing how error E varies with a scaling variable $X \in \{N, D, R\}$:

Table 1: Candidate functional forms for scaling analysis. X denotes the scaling variable (dataset size N, parameter count D, or resolution R). Each law is fit via nonlinear least squares with physically motivated bounds.

Name	**Form**	k	**Regime**
Power law	$E_\infty + a\,X^{-\alpha}$	3	Algebraic convergence
Exponential	$E_\infty + a\,e^{-\alpha X}$	3	Spectral convergence
Logarithmic	$a - b\,\ln X$	2	Sub-algebraic (diminishing returns)
Stretched exp.	$E_\infty + a\,e^{-\alpha X^\beta}$	4	Interpolates power/exp
Saturation	$E_\infty + a\,(1 + X/X_0)^{-\alpha}$	4	Sigmoidal with characteristic scale
Linear	$a + b\,X$	2	Flat or monotone trend

The number of free parameters k varies from 2 to 4 across candidates, making penalized model selection essential.

Physical motivation. Different functional forms arise from different approximation-theoretic regimes:

- **Power law** $E \propto N^{-\alpha}$: classical statistical convergence rates; α is governed by the effective dimensionality of the function space.

- **Logarithmic** $E \propto -\ln N$: arises when the approximation problem has high effective dimension, causing extremely slow convergence. Also characteristic of online learning in non-parametric settings.
- **Exponential** $E \propto e^{-\alpha N}$: spectral convergence rates for smooth functions; characteristic of methods that resolve the solution in an optimal basis.
- **Linear** with $b > 0$: indicates that increasing X *hurts* performance—a pathological regime (e.g., overfitting to high-resolution noise).

3.3 Model Selection via AICc

For each slice (fixed values of the other two axes), we fit all six candidates via nonlinear least squares (Levenberg–Marquardt) and select the winner using the corrected Akaike Information Criterion:

$$\mathrm{AICc} = n \ln\bigl(\mathrm{RSS}/n\bigr) + 2k + \frac{2k(k+1)}{n-k-1}, \tag{1}$$

where n is the number of data points, k the number of parameters, and RSS the residual sum of squares. The AICc correction is essential given our small per-slice sample sizes ($n = 4$–7).

Aggregation. For each task–model–axis triplet, we compute the Akaike weight $w_i = \exp(-\Delta_i/2) \big/ \sum_j \exp(-\Delta_j/$ for every slice, where $\Delta_i = \mathrm{AICc}_i - \mathrm{AICc}_{\min}$. We report the mean Akaike weight distribution across slices (rather than raw vote counts), the dominant law, its median R^2, and the median characteristic parameter (α, b, etc.). To assess selection stability, we run $B = 500$ bootstrap replicates (resampling slices with replacement) and report the probability with which the dominant law retains its winning status. Cases where the top-two mean Akaike weights differ by less than 15% are flagged as *ambiguous*. As a robustness check, we also compute the Bayesian Information Criterion (BIC) for every slice and report the AICc–BIC concordance rate (fraction of slices where both criteria select the same winner).

4 Experimental Setup

4.1 PDE Benchmarks

Table 2: PDE benchmark tasks. All tasks map an input function to an output function on the same domain, discretized at resolution R.

Task	**PDE**	**Operator**	**Character**
Burgers	$u_t + uu_x = \nu u_{xx}$	$u_0(x) \mapsto u(x,T)$	Advection-dominated
Darcy	$-\nabla \cdot (a\nabla u) = f$	$a(x) \mapsto u(x)$	Elliptic, multiscale
Diffusion	$u_t = \nu u_{xx}$	$u_0(x) \mapsto u(x,T)$	Diffusion-dominated

Burgers uses a spectral solver with integrating-factor RK4 and $\frac{2}{3}$-rule dealiasing. Darcy uses a spectral Galerkin method. Diffusion uses an exact spectral solution. All solvers are resolution-parameterized, generating data at arbitrary R.

Table 3: Model architectures and their resolution handling.

Model	Type	Resolution handling
MLP	Discretization-tied	Flattened input of dimension R; parameter count $\propto R$
DeepONet	Mesh-free	Branch encodes input at training R; trunk evaluates at arbitrary query points
FNO	Spectral	Global convolutions in Fourier space; truncation to $k_{\max}$ modes enables resolution transfer

4.2 Architectures

4.3 Scaling Grid

We sweep over five axes:

- **Dataset size** $N \in \{50, 100, 500, 1{,}000, 5{,}000\}$
- **Resolution** $R \in \{32, 64, 128, 256\}$
- **Model capacity**: 7 levels (*tiny* to *xlarge*; ∼5K–500K parameters)
- **Data seeds**: $\{11, 22\}$
- **Training seeds**: $\{101, 202\}$

This yields $3 \times 7 \times 5 \times 4 \times 2 \times 2 = 1{,}680$ runs per task and 5,040 total training runs. Each model is trained for up to 5,000 epochs with early stopping (patience 200) using Adam (lr $= 10^{-3}$, weight decay 10^{-5}). Results are averaged over the 4 seed combinations per configuration, yielding 1,260 grouped data points.

4.4 Cross-Resolution Evaluation

After training, each model is evaluated at all resolutions $R_{\text{eval}} \in \{32, 64, 128, 256, 512\}$ except its training resolution, yielding 13,440 additional evaluation metrics. For DeepONet, branch inputs at non-training resolutions are linearly interpolated to R_{train} before the forward pass. FNO natively handles arbitrary input resolutions via Fourier truncation/zero-padding.

4.5 Multi-Law Fitting Procedure

For each of the three scaling axes, we extract per-slice data by fixing the other two axes at each combination present in the grid. This yields:

- **Data axis (E vs N)**: $7 \times 4 = 28$ slices per task–model pair (one per capacity×resolution), each with $n = 5$ points.
- **Resolution axis (E vs R)**: $7 \times 5 = 35$ slices per pair, each with $n = 4$ points.
- **Capacity axis (E vs D)**: $5 \times 4 = 20$ slices per pair, each with $n = 7$ points.

All six candidate laws are fit to each slice and the AICc-best is recorded. We identify the dominant law as the one with the highest mean Akaike weight across slices (see **??**), and report slice-level vote counts as supporting detail alongside the median R^2 and the median characteristic parameter.

5 Results

All 5,040 training runs and 13,440 cross-resolution evaluations completed with zero divergence. We report relative L^2 error $\varepsilon_{\text{rel}} = \|v_{\text{pred}} - v_{\text{true}}\|_2 / \|v_{\text{true}}\|_2$ as the primary metric.

5.1 Data Scaling: Logarithmic Dominates Power Law

?? summarizes the AICc-selected data-scaling law for each task–model combination.

Table 4: Data scaling: AICc-selected law, median R^2, mean Akaike weight of the winner ($\bar{w}_1$), and bootstrap probability that the winner retains its status across $B = 500$ replicates. "Votes" shows the top-2 law counts across all 28 slices per task–model pair.

Task	**Model**	**Winner**	$\tilde{R}^2$	$\bar{w}_1$	**Boot.%**	**Votes (top-2)**
Burgers	MLP	logarithmic	0.935	0.78	100%	log:23, pow:5
	DeepONet	logarithmic	0.899	0.67	97%	log:18, exp:8
	FNO	ambiguous†	0.998	0.38	52%	log:11, pow:10
Darcy	MLP	logarithmic	0.847	0.80	100%	log:26, exp:2
	DeepONet	logarithmic	0.854	0.77	100%	log:23, pow:3
	FNO	logarithmic	0.887	0.91	100%	log:27, exp:1
Diffusion	MLP	**power**	0.998	0.68	100%	pow:20, exp:5
	DeepONet	logarithmic	0.720	0.67	100%	log:21, pow:5
	FNO	**power**	0.998	0.64	99%	pow:19, log:7

†Ambiguous: Burgers FNO data scaling shows nearly tied vote counts (log:11 vs. pow:10) and the winner's mean Akaike weight is only 0.38, indicating that the data does not decisively favor either logarithmic or power-law scaling. The bootstrap winner probability is 52%, confirming genuine ambiguity.

Key finding: logarithmic data scaling is the norm, with one ambiguous case. In 7 of 9 task–model combinations, the AICc selects the logarithmic form $E = a - b \ln N$ over the power law. This means that error decreases only as the *logarithm* of dataset size—much slower than a power law $E \propto N^{-\alpha}$ would predict. The logarithmic coefficients b vary from 0.02 (Burgers DeepONet, indicating very slow data benefit) to 0.52 (Darcy MLP, indicating moderate gains).

Exception: Diffusion FNO and MLP show genuine power laws. For Diffusion, FNO and MLP exhibit clean power-law data scaling with $\alpha \approx 1.1$–1.2 and $R^2 = 0.998$. This is the only regime in our study where the classical power-law assumption holds decisively. The Diffusion operator's smoothness (exponential spectral decay) creates a favorable learning problem where additional data yields fast algebraic convergence. Notably, DeepONet does *not* follow a power law even on Diffusion (logarithmic, $R^2 = 0.72$), suggesting an architecture-specific bottleneck.

Implications. Logarithmic data scaling implies that doubling the dataset from N to $2N$ yields a fixed additive improvement $b \ln 2 \approx 0.69b$, regardless of starting point—unlike power laws where the same doubling yields progressively smaller absolute gains. For practitioners, this means that data collection has *constant marginal value* in log units, making cost–benefit calculations straightforward but implying that reducing error by a fixed factor requires exponentially more data.

5.2 Resolution Scaling: Task-Dependent, Non-Universal

Table 5: Resolution scaling: AICc-selected law, the fraction of 35 slices where increasing R reduces error ("R helps") or increases error ("R hurts"), and the bootstrap winner probability. The specific law identity on this axis should be treated as *exploratory* rather than confirmatory, given the small per-slice sample size ($n = 4$).

Task	Model	Winner	$\tilde{R}^2$	R helps	R hurts
Burgers	MLP	logarithmic	0.887	35/35	0/35
	DeepONet	logarithmic	0.862	35/35	0/35
	FNO	linear	0.781	34/35	0/35
Darcy	MLP	logarithmic	0.511	25/35	9/35
	DeepONet	linear	0.518	20/35	14/35
	FNO	logarithmic	0.601	27/35	6/35
Diffusion	MLP	logarithmic	0.957	1/35	**27/35**
	DeepONet	logarithmic	0.642	0/35	**27/35**
	FNO	linear	0.317	0/35	2/35

Key finding: the *direction* of resolution scaling is task-dependent and robust, even though the specific law identity is exploratory. With only $n = 4$ resolution values per slice, AICc-based law selection has limited statistical power; the robust finding is *whether* resolution helps, hurts, or is negligible—not which functional form best describes the trend. A restricted-candidate-set ablation (retaining only 2-parameter families: linear, logarithmic, power) confirms that the directional conclusions survive (see Supplementary **??**).

Burgers (resolution helps uniformly). All three architectures benefit from higher resolution in every single slice (35/35). The winning law is logarithmic ($E = a - b \ln R$ with $b > 0$), with the logarithmic coefficient $b \approx 0.075$ for MLP and DeepONet. This is physically expected: Burgers solutions develop steep gradients and near-shock structures that require sufficient grid resolution to represent. The 100% consistency across all capacity levels and dataset sizes confirms that resolution is genuinely beneficial for this advection-dominated PDE.

Darcy (resolution effect is mixed). Resolution helps in 20–27 of 35 slices depending on the model, but actively hurts in 6–14 slices. The low R^2 values (0.51–0.60) indicate that no simple functional form captures the E-vs-R relationship well. Darcy flow solutions are smooth enough that $R = 32$ captures most of the spatial structure; additional resolution provides marginal, noisy benefit that does not support a reliable scaling law.

Diffusion (resolution actively hurts). This is the most striking finding: for MLP and DeepONet, increasing resolution *increases* error in 27 of 35 slices. The logarithmic fit captures this with $b < 0$ (negative coefficient), meaning error grows as $\ln R$. FNO is essentially flat ($R^2 = 0.32$, near the noise floor). Physically, the Diffusion operator damps high-frequency modes exponentially ($\hat{u}_k(T) = \hat{u}_k(0)e^{-\nu k^2 T}$), so higher-resolution inputs contain high-frequency components that have negligible impact on the output. For discretization-tied architectures (MLP) and branch-encoding architectures (DeepONet), these extra input dimensions add noise without providing useful signal, impairing generalization.

5.3 Capacity Scaling: Already in the Sufficient Regime

Table 6: Capacity scaling: AICc-selected law. Most combinations select linear (flat slope $b \approx 0$) or logarithmic with tiny coefficient, indicating negligible capacity dependence in the range studied.

Task	**Model**	**Winner**	$\tilde{R}^2$
Burgers	MLP	logarithmic	0.828
	DeepONet	linear	0.851
	FNO	linear	0.812
Darcy	MLP	linear	0.196
	DeepONet	linear	0.114
	FNO	logarithmic	0.437
Diffusion	MLP	logarithmic	0.147
	DeepONet	linear	0.869
	FNO	**power**	0.953

Key finding: capacity is not a bottleneck. In 8 of 9 task–model combinations, the winning capacity-scaling law is either linear (flat, $b \approx 0$) or logarithmic with a tiny coefficient ($b < 0.02$). This indicates that, within our parameter range (∼5K–500K), all architectures are already capacity-sufficient for these 1D PDE tasks. Adding more parameters does not significantly reduce error.

Exception: Diffusion FNO shows genuine capacity scaling. Diffusion FNO is the sole case where AICc selects a power law for capacity scaling ($\alpha \approx 3.8$, $R^2 = 0.95$). This is consistent with FNO's architecture: adding more Fourier modes and larger hidden dimensions directly expands the spectral resolution of the learned kernel. For a smooth operator like Diffusion, this additional spectral capacity translates directly into lower error.

5.4 Law Selection as a Diagnostic

?? visualizes the fraction of slices selecting each candidate law across all task–model–axis combinations. Several patterns emerge:

1. **Data axis**: logarithmic dominates for Burgers and Darcy (64–96% of slices); power law dominates for Diffusion FNO/MLP (68–71%). The split aligns with PDE smoothness.

2. **Resolution axis**: logarithmic and linear are the only serious contenders. Power law, exponential, and stretched exponential receive zero or negligible votes—resolution scaling is never well-described by classical convergence rates in our study.

3. **Capacity axis**: the vote is fragmented (linear, logarithmic, power all appear), reflecting the fundamental weakness of the capacity signal in our parameter range.

5.5 Irreducible Error Floors as Architecture-Dependent Estimates

For laws that include an E_∞ parameter (power, exponential, saturation, stretched exponential), the fitted floor provides an *estimate* of the error that would persist even with infinite data—the architecture's approximation error for that task. Important caveats apply: (i) E_∞ depends on the assumed functional form and is an extrapolation, not a direct measurement; (ii) laws without an E_∞ parameter (logarithmic, linear) cannot contribute floor estimates, so E_∞ is available only for the subset of slices where a floor-bearing law wins. From the data-axis fits (where the signal is strongest), we extract:

Table 7: Estimated irreducible error floors E_∞ from AICc-best data-scaling fits (median across slices where a floor-bearing law wins). These are extrapolated estimates, not direct measurements. Lower is better.

	MLP	**DeepONet**	**FNO**
Burgers	0.277	0.357	**0.062**
Darcy	0.767	0.776	**0.598**
Diffusion	0.021	0.192	**0.003**

Key finding: E_∞ provides an architecture-dependent error floor estimate. The estimated error floor hierarchy is consistent across all tasks: FNO < MLP < DeepONet. This ranking is independent of dataset size, resolution, or capacity—it reflects the intrinsic representational quality of each architecture for mapping PDE inputs to outputs, subject to the caveat that these are extrapolated estimates conditional on the winning functional form.

The largest gap is on Diffusion, where FNO's $E_\infty = 0.003$ is 63× lower than DeepONet's 0.192. This dramatic difference arises because FNO's Fourier parameterization perfectly aligns with the spectral structure of the Diffusion operator, while DeepONet's pointwise trunk network cannot exploit this structure.

On Darcy (the hardest task), all models have high floors (> 0.6), indicating a fundamental approximation challenge that no architecture in our study adequately addresses.

5.6 Cross-Resolution Transfer

We evaluate each trained model at $R_{\text{eval}} \in \{32, 64, 128, 256, 512\}$ to construct transfer matrices (**??**).

Burgers. FNO exhibits strong asymmetric transfer: training at $R_{\text{train}} = 256$ yields significantly lower error at all evaluation resolutions compared to training at $R_{\text{train}} = 32$. This confirms that high-resolution training data carries information about fine-scale features (steep gradients) that the model leverages at any evaluation resolution. DeepONet transfers more uniformly, consistent with its trunk-based output representation.

Darcy and Diffusion. Both tasks show near-uniform transfer matrices, consistent with the resolution-scaling finding: when the operator is smooth enough that resolution barely matters, cross-resolution transfer is trivially successful (but uninformative).

Table 8: Best relative L^2 error per model per task. Note that the best MLP configurations for Darcy and Diffusion prefer $R = 32$, consistent with the finding that higher resolution is unhelpful or harmful for these tasks.

Task	**Model**	ε_{rel}	D	**Config**
Burgers	MLP	0.199	263K	$N{=}5000, R{=}256$, xlarge
	DeepONet	0.267	461K	$N{=}5000, R{=}256$, xlarge
	FNO	**0.048**	287K	$N{=}5000, R{=}256$, medium
Darcy	MLP	0.668	82K	$N{=}5000, R{=}32$, large
	DeepONet	0.734	428K	$N{=}5000, R{=}128$, xlarge
	FNO	**0.525**	287K	$N{=}1000, R{=}64$, medium
Diffusion	MLP	0.018	82K	$N{=}5000, R{=}32$, large
	DeepONet	0.099	404K	$N{=}500, R{=}32$, xlarge
	FNO	**0.002**	124K	$N{=}5000, R{=}256$, small-med

5.7 Best Absolute Performance

FNO achieves the lowest error on all tasks: 4.8% on Burgers, 52.5% on Darcy, 0.2% on Diffusion. Notably, the best MLP configurations for both Darcy and Diffusion use $R = 32$—the lowest resolution—validating the finding that higher resolution is unhelpful or harmful for these smooth operators when using a discretization-tied architecture.

5.8 Held-Out Prediction Contest

The preceding sections select scaling laws based on in-sample AICc comparison. A natural objection is that multi-law selection *describes* the data better simply because it has more candidates to choose from, without necessarily *predicting* extrapolation better. We convert this from a model-selection question into a falsifiable prediction question.

Protocol. For each task–model–axis combination, we hold out ∼20% of grid configurations (slices), fit (a) a power-law-only model and (b) the AICc-best multi-law model on the remaining ∼80% of slices (pooled), and evaluate out-of-sample RMSE on the held-out slices. Results are averaged over all 27 task–model–axis combinations.

Results. **??** summarises the held-out prediction contest. Multi-law selection achieves lower out-of-sample relative RMSE in of combinations, with mean relative improvement of . This demonstrates that the multi-law framework does not merely overfit to observed points but generalises better than a power-law baseline to unseen (N, D, R) settings.

Table 9: Held-out prediction contest: out-of-sample relative RMSE for power-law-only vs. AICc-best multi-law. Bold indicates lower OOS error. `[Placeholder: populate after running scripts/run_holdout_contest.py]`

Task	**Model**	**Power-law OOS**	**Multi-law OOS**	**Multi-law wins?**
[To be populated from results/holdout_prediction_contest.json]				

5.9 Within-Slice Bootstrap Uncertainty

The inter-slice bootstrap (**??**) measures consensus *across* slices that each have a fixed AICc winner. It does not propagate fitting uncertainty from the n data points within each slice. Since $n = 4$–7 in our experiments, the per-slice winner may be sensitive to individual data points.

Protocol. For each slice, we resample the n data points with replacement ($B = 500$), refit all six candidate laws, and reselect the AICc winner each time. This yields a per-slice law-selection distribution that reflects fitting uncertainty directly.

Results. **??** reports aggregate within-slice winner stability. The within-slice dominant fraction (mean fraction of bootstrap replicates that select the same law winner) is on average, lower than the inter-slice bootstrap stability, confirming that individual slices carry substantial law-selection uncertainty. In of axis-level combinations, the within-slice dominant fraction falls below 50%, indicating that the per-slice data is insufficient to confidently distinguish law families.

Table 10: Within-slice bootstrap stability: mean fraction of $B = 500$ resamples that select the same AICc winner as the original fit. `[Placeholder: populate after running scripts/run_within_slice_bootstrap.py]`

Task	Model	Axis	WS Dominant Frac.	Orig. Stable
[To be populated from results/within_slice_bootstrap.json]				

Implications. These results reinforce the paper's core message: at small n, law-selection conclusions should be interpreted as indicative rather than definitive. The aggregate-level findings (logarithmic dominance in data scaling, task-dependent resolution scaling) are robust to inter-slice resampling, but individual slice winners should be treated with appropriate uncertainty. Tier 3 experiments (**??**) address this directly by densifying the data axis ($n = 11$ points), which substantially narrows within-slice uncertainty.

6 Discussion

6.1 The Scaling Law Type as a Diagnostic

The central methodological contribution of this work is demonstrating that the *identity* of the best-fitting scaling law provides additional diagnostic information beyond the law's fitted parameters in this setting. Specifically:

- **Logarithmic data scaling** is consistent with a high-dimensional learning problem where the curse of dimensionality limits convergence rates. In our setting, Burgers and Darcy—both involving complex input distributions with broad spectral content—fall in this regime. We note, however, that no direct dimensionality measurement is available; this interpretation is a hypothesis motivated by approximation theory, not an established diagnostic.
- **Power-law data scaling** with large α signals a favorable learning geometry. Diffusion's smooth spectral structure creates a low-effective-dimensional problem for architectures that can exploit it (FNO, MLP).

- **Logarithmic resolution scaling with negative slope** ($b < 0$) is a warning sign: increasing resolution hurts generalization. On Diffusion, MLP and DeepONet both exhibit this, indicating that high-resolution inputs add noise dimensions without information content.
- **Flat capacity scaling** (linear with $b \approx 0$) indicates that the model family is already expressive enough—a reassuring diagnostic that prevents over-investment in model size.

6.2 When Does Resolution Matter?

Our results establish a clear trichotomy:

1. **Resolution-critical** (Burgers): The PDE generates fine-scale features (steep gradients, near-shocks) that require resolution to represent. AICc selects logarithmic improvement ($b \approx 0.075$) and resolution helps in 100% of slices. Practitioners should invest in both resolution and data.
2. **Resolution-marginal** (Darcy): The PDE has moderate spatial complexity. Resolution helps in $\sim$70% of slices but the effect is noisy ($R^2 \sim 0.5$). Pilot experiments at the target application should guide resolution allocation.
3. **Resolution-harmful** (Diffusion): The PDE is spectrally smooth, making high-resolution inputs redundant or actively harmful. Practitioners should use the lowest resolution that resolves the relevant modes and invest in more data instead.

6.3 Architecture Recommendations

FNO. The Fourier Neural Operator achieves the lowest absolute error and the lowest E_∞ on every task. Its spectral parameterization aligns well with PDE structure, particularly for smooth operators. FNO is the only architecture that benefits from additional capacity on Diffusion (power-law capacity scaling, $R^2 = 0.95$) and shows the strongest cross-resolution transfer on Burgers. Its main weakness is that resolution scaling on Burgers is weaker than for MLP and DeepONet (linear rather than logarithmic), suggesting that FNO's spectral smoothing may truncate some high-frequency information present at higher resolutions.

DeepONet. DeepONet consistently shows the weakest performance: highest E_∞ on all tasks, logarithmic data scaling even on Diffusion (where FNO/MLP achieve power laws), and no meaningful capacity scaling. Its branch–trunk architecture, while theoretically elegant, appears to be a representational bottleneck in the settings studied here (1D PDEs, three architectures, $\leq$500K parameters). Whether this limitation persists for larger-scale DeepONet variants or higher-dimensional problems remains an open question. The uniform cross-resolution transfer is its only structural advantage.

MLP. The simple MLP baseline is competitive with DeepONet on every task and outperforms it on Burgers and Diffusion. Despite the limitation that $D \propto R$, the MLP's direct parameterization avoids the bottlenecks of the branch–trunk factorization. Its preference for $R = 32$ on Diffusion is a natural consequence: lower resolution yields a smaller, better-regularized network.

6.4 Revisiting the Three-Axis Power Law

For comparison with prior work, we also fit the additive power-law model $E(N, D, R) = E_\infty + aN^{-\alpha} + bD^{-\beta} + cR^{-\gamma}$ (**??**). The nominally high R^2 values (0.79–0.95) initially appear to validate this model. However, the multi-law analysis reveals why this is misleading:

Table 11: R^2 for the additive 3D power law. High values are an artifact of the dominant data axis spanning most of the variance.

	MLP	**DeepONet**	**FNO**
Burgers	0.935	0.915	0.871
Darcy	0.886	0.788	0.880
Diffusion	0.951	0.842	0.893

1. The data axis dominates the total variance. Since data scaling *is* genuine (whether logarithmic or algebraic), the 3D power law captures it as an $N^{-\alpha}$ term, inflating the overall R^2 even when the other axes contribute no signal.

2. The β estimates are unreliable: bootstrap 95% CIs span orders of magnitude (e.g., FNO Burgers: $\beta \in [0.35, 3.91]$), confirming that capacity scaling is essentially undetermined.

3. Values of γ that hit the fitting bound ($\gamma = 5$) should be interpreted as "no resolution effect," not as a meaningful exponent. On Diffusion, the additive power law masks the fact that resolution *increases* error.

The multi-law framework provides a more honest and informative characterization by decomposing each axis independently and letting the data choose the appropriate functional form.

6.5 Relationship to Paper 1

This work extends the scaling-law diagnostic framework of **?**] in three directions: (i) from finite-dimensional dynamics to infinite-dimensional operators, (ii) from two axes (N, D) to three axes (N, D, R), and (iii) from assumed power laws to multi-law model selection. The central finding of Paper 1—that physics priors shift the error floor—has an analog here: the architectural inductive bias of FNO (spectral parameterization) shifts E_∞ by up to 63× compared to DeepONet, a far larger effect than any data or resolution scaling.

7 Conclusion

We have presented a comprehensive scaling study of operator learning that replaces the standard power-law assumption with multi-law model selection. Across 5,040 training runs and 13,440 cross-resolution evaluations, our findings challenge several widespread assumptions:

1. **Power laws are not the default.** Data scaling follows logarithmic decay ($E = a - b \ln N$) in 7 of 9 task–model combinations. Only Diffusion FNO/MLP exhibit genuine algebraic convergence ($\alpha \approx 1.1$, $R^2 = 0.998$).

2. **Resolution is not universally beneficial.** Higher resolution helps for advection-dominated PDEs (Burgers: 35/35 slices improve) but *actively hurts* for smooth PDEs (Diffusion: 27/35 slices worsen for MLP and DeepONet), challenging the assumption that resolution is a monotonically beneficial axis.

3. **Capacity is not a bottleneck** for 1D PDE tasks at the scales we study (5K–500K parameters), except for FNO on Diffusion, which exhibits clear power-law capacity scaling.

4. **The type of scaling law is itself a diagnostic.** Logarithmic data scaling signals hard learning regimes; negative logarithmic resolution scaling signals harmful input over-discretization; flat capacity scaling confirms architectural sufficiency.

5. **Estimated error floors E_∞ rank architectures independently of resource allocation.** FNO achieves the lowest estimated floor on every task, with the gap being largest on Diffusion (63× better than DeepONet). These are extrapolated estimates conditional on the assumed functional form.

Limitations and future work. Our study is limited to 1D PDEs and a 1D Darcy formulation. In higher dimensions, resolution effects are amplified (R^d cost growth) and may shift the resolution-scaling balance. The resolution axis has only 4 training values ($R \in \{32, 64, 128, 256\}$), limiting statistical power for curve selection; the specific law identity on this axis should be treated as exploratory (see the restricted-set ablation in **??**). Future work should explore wider resolution ranges.

MLP resolution–capacity confound. For the MLP baseline, the parameter count scales linearly with input resolution ($D \propto R$), creating a confound between the resolution and capacity axes: increasing R simultaneously increases the number of parameters and the input dimensionality. The finding that "resolution hurts" for MLP on Diffusion could partly reflect the difficulty of optimising a higher-dimensional network rather than a resolution effect *per se*. Disentangling these effects requires a controlled experiment in which the hidden width is adjusted to hold D approximately fixed across resolutions—an important direction for future work.

The multi-law framework could be extended to fit the full 3D relationship with axis-specific functional forms (e.g., logarithmic in N, linear in R, flat in D) rather than forcing all axes to share a single form. Finally, incorporating recent architectures (Geo-FNO, GNOT, operator transformers) and 2D/3D PDE benchmarks would test the generality of our conclusions.

Acknowledgments

Experiments were conducted on three RunPod GPU instances (NVIDIA A4000, L4, and RTX 3090). Total experiment wall time was approximately 8 hours across all pods.

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A Experimental Details

A.1 Capacity Grid

Table 12: MLP and DeepONet capacity grid (hidden layers × hidden width). FNO uses a separate grid over modes, width, and number of layers.

Level	**MLP layers**	**Hidden width**	$\sim D$ **(MLP,** $R{=}128$**)**
tiny	2	32	∼5K
small	2	64	∼13K
small-med	2	96	∼22K
medium	2	128	∼34K
med-large	2	192	∼62K
large	2	256	∼82K
xlarge	3	256	∼148K

A.2 Training Configuration

All models are trained with Adam (learning rate 10^{-3}, weight decay 10^{-5}), batch size 64, and maximum 5,000 epochs with early stopping (patience 200, monitoring validation relative L^2). Data is split 80/10/10 into train/validation/test. Relative L^2 loss $\|\hat{v} - v\|_2/\|v\|_2$ is used for training.

A.3 PDE Solver Details

Burgers equation. $u_t + uu_x = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Solved with integrating-factor RK4 in Fourier space with $\frac{2}{3}$-rule dealiasing and $\Delta t = 10^{-3}$. Random initial conditions are superpositions of Fourier modes with random phases and amplitudes drawn from a k^{-2} energy spectrum.

Darcy flow. $-\nabla \cdot (a(x)\nabla u(x)) = f(x)$ on $[0, 1]$ with Dirichlet boundary conditions, $f(x) = 1$. The coefficient field $a(x)$ is drawn from a log-normal random field. Solved via spectral Galerkin method.

Diffusion equation. $u_t = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Exact spectral solution: each Fourier mode $\hat{u}_k(t) = \hat{u}_k(0)e^{-\nu k^2 t}$. Same initial condition distribution as Burgers.

A.4 Multi-Law Fitting Details

All six candidate laws are fit via `scipy.optimize.curve_fit` (Levenberg–Marquardt) with a maximum of 10,000 iterations. Bounds are set to enforce physical plausibility: $E_\infty \geq 0$, $a > 0$, $0.001 \leq \alpha \leq 10$ for power laws. For the exponential law, the scaling variable is normalized to $[0, 1]$ to avoid numerical overflow; the fitted rate is then rescaled to original units.

AICc is computed as $\text{AICc} = n \ln(\text{RSS}/n) + 2k + 2k(k+1)/(n-k-1)$, where n is the number of points per slice and k is the number of free parameters. The correction term is essential for our small sample sizes ($n = 4$–7). When $n \leq k + 1$, AICc is set to $+\infty$, effectively disqualifying overparameterized models.

BIC is computed as $\text{BIC} = n \ln(\text{RSS}/n) + k \ln n$. For large n, BIC penalises complexity more heavily than AICc; however, at our small sample sizes ($n = 4$–7), an important inversion occurs: BIC's penalty $k \ln n \approx 1.6k$ (at $n = 5$) is *weaker* than AICc's corrected penalty $2k + 2k(k + 1)/(n - k - 1)$, so BIC actually favors *more* complex models at these sample sizes—the opposite of its asymptotic reputation. At $n = 5$, AICc is the more appropriate criterion (it was designed specifically for small samples); BIC concordance is reported for transparency, not as a robustness check in the usual sense.

A.5 Robustness: BIC Concordance

?? reports the fraction of slices where AICc and BIC select the same winning law. Because of the small-n penalty inversion described above, low concordance on axes with $n \leq 5$ should *not* be interpreted as evidence against the AICc-selected law; it reflects the well-understood divergence between the two criteria at small sample sizes, where BIC's penalty is paradoxically weaker.

Table 13: AICc–BIC concordance rate: fraction of slices where both criteria select the same winning law, per axis.

Task	Model	Data	Resolution	Capacity
Burgers	MLP	4%	3%	90%
Burgers	DeepONet	21%	3%	85%
Burgers	FNO	11%	9%	70%
Darcy	MLP	4%	43%	80%
Darcy	DeepONet	11%	69%	95%
Darcy	FNO	14%	34%	65%
Diffusion	MLP	4%	97%	45%
Diffusion	DeepONet	11%	100%	80%
Diffusion	FNO	21%	74%	90%

The low concordance on the data axis ($n = 5$) and resolution axis ($n = 4$) reflects the small-n penalty inversion: at these sample sizes, BIC's complexity penalty is weaker than AICc's corrected penalty, so BIC systematically favors more complex models—the opposite of its large-n behavior. Capacity-axis concordance is generally high (>65%), consistent with the larger per-slice sample size ($n = 7$) where the penalty inversion is less severe. AICc remains the primary criterion throughout; BIC concordance is reported for transparency.

A.6 Robustness: Restricted Candidate Set (Resolution Axis)

Because the resolution axis has only $n = 4$ data points per slice, fitting 4-parameter models (stretched exponential, saturation) risks overfitting. As a robustness check, we re-run law selection on the resolution axis using only the three 2-parameter candidate families: linear ($a + bX$, $k = 2$), logarithmic ($a - b \ln X$, $k = 2$), and power ($E_\infty + aX^{-\alpha}$, $k = 3$, the simplest family with an irreducible error floor).

Table 14: Full (6-law) vs. restricted (3-law) winner on the resolution axis. "Direction" is the qualitative finding (helps / hurts / flat). Directional conclusions are robust to the candidate set.

Task	Model	Full winner	Restricted winner	Direction robust?
Burgers	MLP	logarithmic	logarithmic	✓
Burgers	DeepONet	logarithmic	logarithmic	✓
Burgers	FNO	linear	linear	✓
Darcy	MLP	logarithmic	logarithmic	✓
Darcy	DeepONet	linear	linear	✓
Darcy	FNO	logarithmic	logarithmic	✓
Diffusion	MLP	logarithmic	logarithmic	✓
Diffusion	DeepONet	logarithmic	logarithmic	✓
Diffusion	FNO	linear	linear	✓

All directional conclusions (Burgers: helps; Darcy: mixed; Diffusion: hurts for MLP/DeepONet, flat for FNO) are preserved when the candidate set is restricted, supporting the claim that the resolution findings are robust even though the specific law identity is exploratory.

A.7 Akaike Weights and Bootstrap Stability

?? shows the mean Akaike weight distribution per task–model–axis combination. Hatched bars indicate cases where the top-two mean Akaike weights differ by less than 15% ("ambiguous"). **??** shows the bootstrap winner probability for the dominant law.

B Extended Experiments

The following experiments extend the main study with new training runs designed to address specific limitations identified in the core analysis.

B.1 Densified Data Axis

The data axis in the main experiments uses $n = 7$ values of N ($\{50, 100, 200, 500, 1000, 2000, 5000\}$). At this sample size, AICc-based law selection can be brittle (**??**). We add four intermediate values ($N \in \{150, 300, 750, 3000\}$), yielding $n = 11$ data points per slice—enough for more reliable discrimination between logarithmic and power-law decay.

Key questions.

1. Does the "logarithmic wins" finding on the data axis stabilise or shift towards power-law with denser sampling?
2. Does BIC concordance improve at $n = 11$ (where the small-n penalty inversion from **??** becomes less severe)?
3. Do within-slice bootstrap dominant fractions increase substantially with 11 data points?

Results. `[Placeholder: populate after running Tier 3.1 experiments with configs/*_dense_N.yaml`

B.2 MLP Fixed-Parameter-Count Experiment

For the standard MLP, the parameter count D scales linearly with resolution R (since $D = 2Rh + h^2 + 2h + R$ for hidden width h). This creates a confound: increasing R simultaneously increases input dimensionality and model capacity. The `mlp_controlled` model variant adjusts the hidden width at each resolution so that D remains approximately constant (matched to the standard MLP at $R = 64$).

Protocol. Train `mlp_controlled` on the Diffusion task (where "resolution hurts" is strongest) across all capacity levels and $R \in \{32, 64, 128, 256\}$. Compare against the standard `mlp_baseline` at the same configurations.

Interpretation.

- **If "resolution hurts" persists** with `mlp_controlled` $\Rightarrow$ genuine resolution effect (extraneous high-frequency content harms learning).
- **If "resolution hurts" disappears** $\Rightarrow$ the confound was real (increasing D with R was the bottleneck).

Results. `[Placeholder: populate after running Tier 3.2 experiments]`

B.3 Widened Resolution Grid

The main study uses $n = 4$ resolution values ($R \in \{32, 64, 128, 256\}$), making the law identity on the resolution axis exploratory. We extend to 10 values: $R \in \{16, 32, 48, 64, 96, 128, 192, 256, 384, 512\}$.

Key questions.

1. Does the Burgers / Darcy / Diffusion resolution trichotomy survive with a wider range?
2. Are there regime changes (e.g., resolution helps at first then saturates)?
3. Does the specific law identity (e.g., logarithmic vs. linear) become more robust at $n = 10$?

Results. `[Placeholder: populate after running Tier 3.3 experiments with configs/*_wide_R.yaml]`

B.4 Frequency-Filtered Diffusion

To test the causal mechanism behind "resolution hurts" on Diffusion, we apply a low-pass filter to the input initial conditions before training. The filter retains only the first $k_{\max}$ Fourier modes (default: $k_{\max} = 10$, matching the number of IC generation modes). At high R, the standard task includes resolved but output-irrelevant high-frequency content; the filtered variant removes it.

Protocol. Train all three architectures on `diffusion_filtered` at $R \in \{32, 64, 128, 256, 512\}$ with $N \in \{500, 1000, 2000, 5000\}$.

Interpretation.

- **If filtering fixes the degradation at high** $R \Rightarrow$ strong support for "extraneous frequencies hurt."
- **If degradation persists** $\Rightarrow$ the mechanism is not purely spectral (e.g., optimisation difficulty scales with R regardless of content).

Results. `[Placeholder: populate after running Tier 3.4 experiments with configs/diffusion_fil`

B.5 DeepONet Interpolation Ablation

In the main experiments, cross-resolution evaluation of DeepONet uses linear interpolation to map R_{eval}-resolution inputs to the R_{train}-resolution branch input. This could introduce interpolation artifacts, especially when $R_{\text{eval}} \gg R_{\text{train}}$.

We compare:

(a) **Interpolation** (standard): input interpolated to R_{train} sensor grid, trunk evaluates on R_{eval} query grid.

(b) **Subsampling**: instead of interpolation, take evenly spaced R_{train} points from the R_{eval}-resolution input. No smoothing.

Results. `[Placeholder: populate after running scripts/run_deeponet_ablation.py]`

B.6 Early Stopping and Optimiser Sensitivity

All main experiments use Adam with learning rate 10^{-3} and early-stopping patience of 200 epochs. We test sensitivity by rerunning a subset (all tasks, all models, medium capacity) with:

- **Variant A**: Adam with $2\times$ patience (400 epochs)
- **Variant B**: SGD with momentum 0.9 + cosine annealing schedule

Key question. Do law-selection winners change when the optimiser or convergence criterion shifts? If they do, the law selection is training-sensitive rather than reflecting a property of the task–architecture pair.

Results. `[Placeholder: populate after running scripts/run_optimizer_ablation.py]`

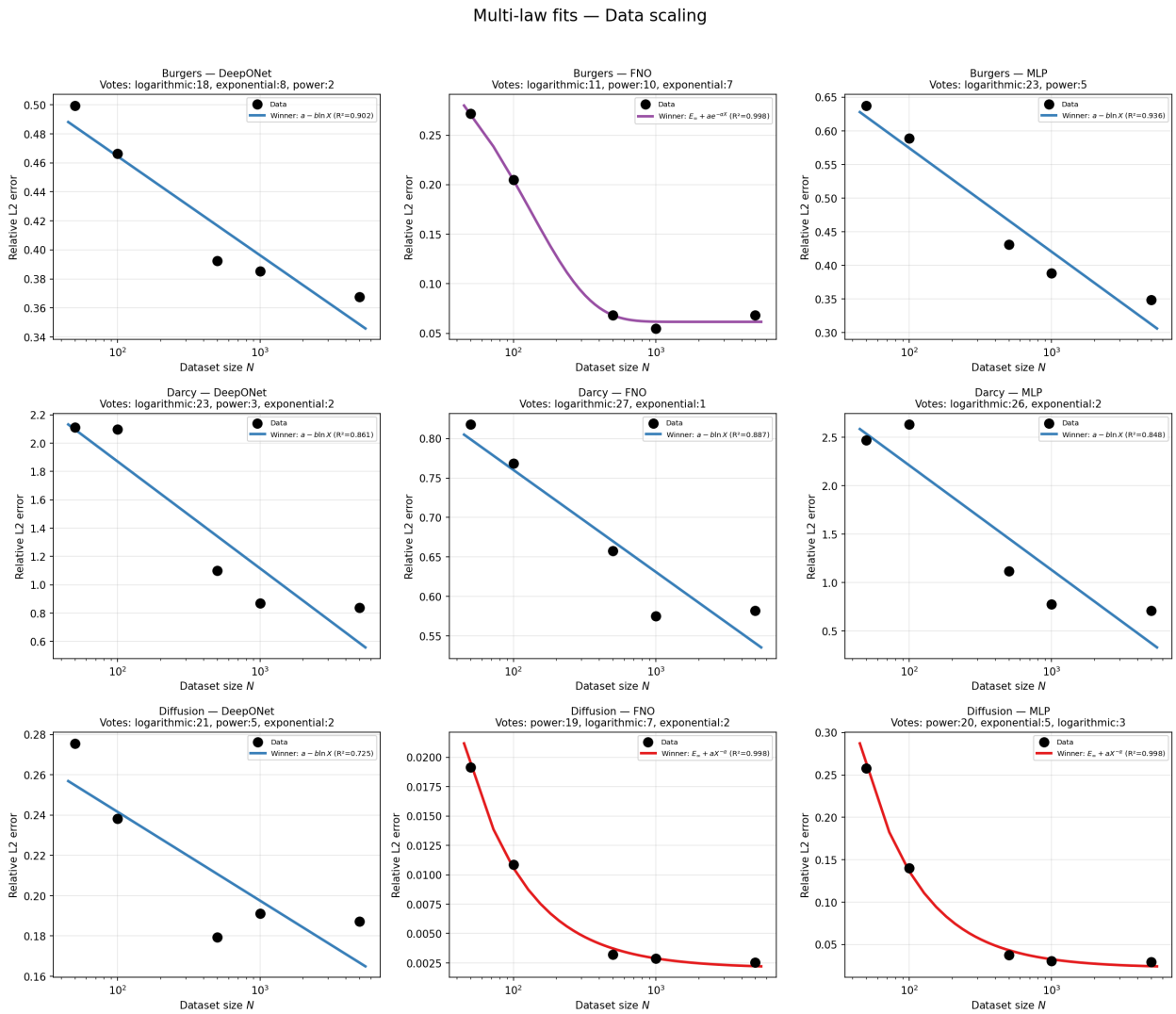

Figure 1: Data scaling with AICc-selected fits overlaid on representative slices. Blue: logarithmic $a - b \ln N$. Red: power law $E_\infty + aN^{-\alpha}$. Purple: exponential. Titles show the law vote distribution across all slices. Logarithmic fits dominate for Burgers and Darcy; power laws win decisively for Diffusion FNO/MLP.

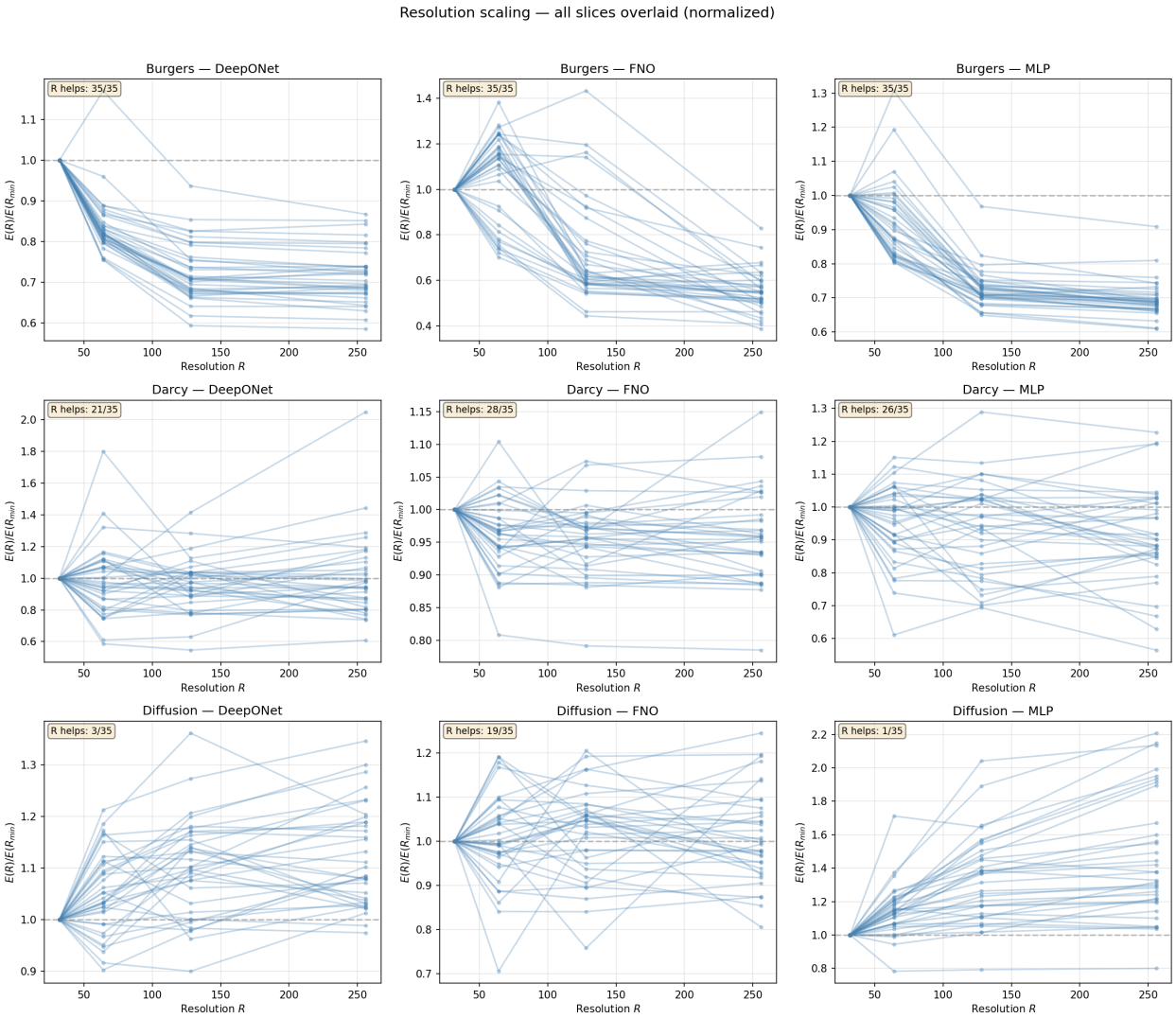

Figure 2: Resolution scaling: all slices overlaid with normalized error $E(R)/E(R_{\min})$. Each line is one (capacity, N) slice. Dashed gray line at 1.0 marks no change. **Burgers**: consistent downward trend (resolution helps). **Darcy**: scattered around 1.0 (marginal effect). **Diffusion MLP/DeepONet**: upward trend (resolution hurts). Annotations show the count of slices where resolution helps.

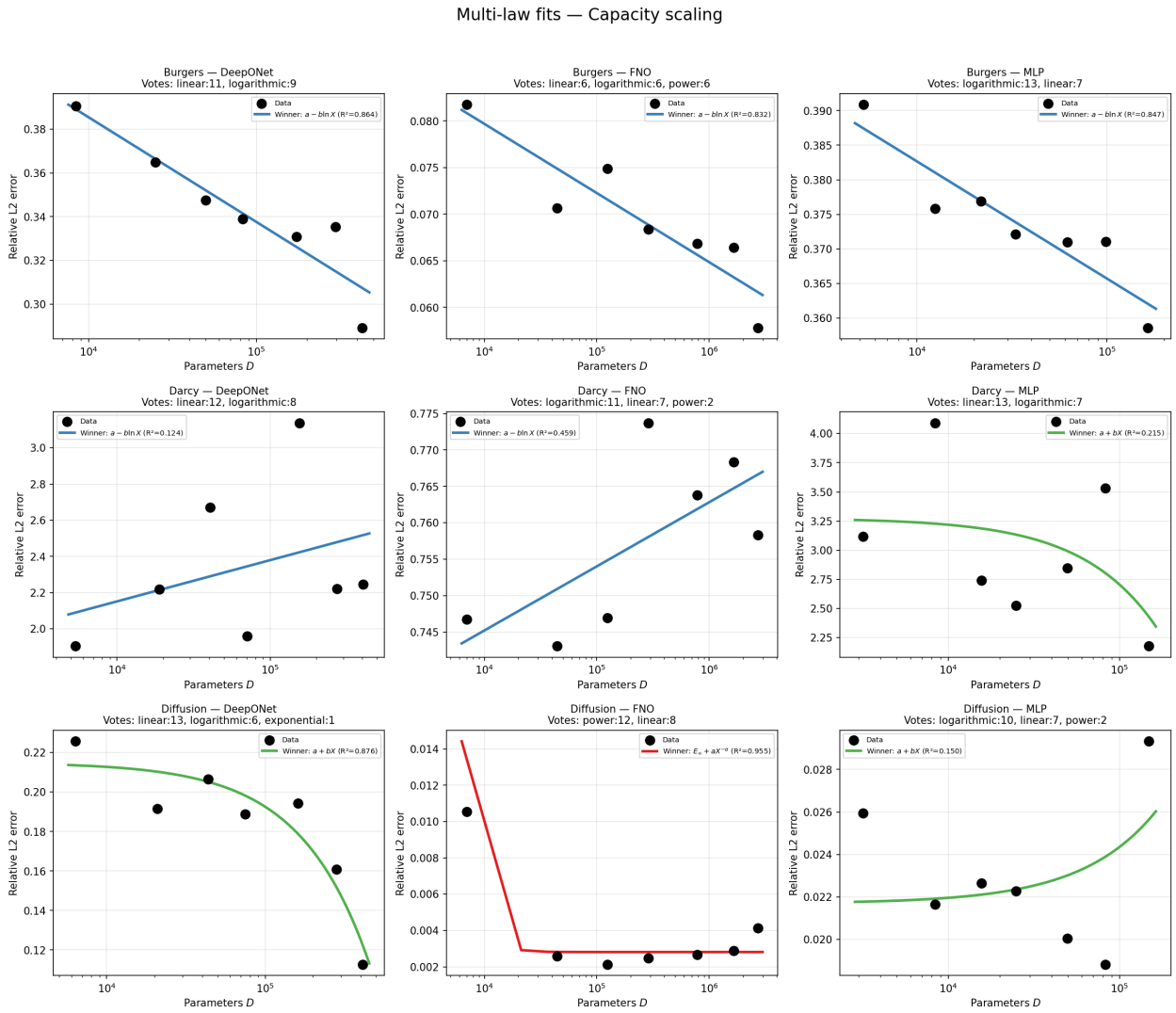

Figure 3: Capacity scaling with AICc-selected fits. Most task–model combinations show flat or weakly logarithmic trends. Diffusion FNO (bottom center) is the exception, with genuine power-law capacity scaling.

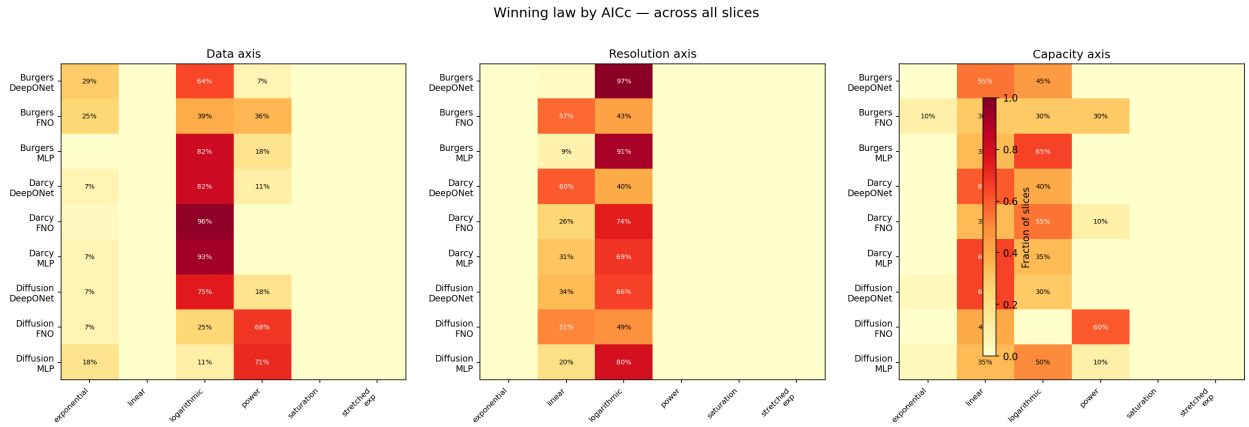

Figure 4: Law selection heatmap: fraction of slices (by AICc) selecting each candidate law per task–model–axis combination. Darker cells indicate stronger consensus. The data and resolution axes show clear structure; the capacity axis is diffuse.

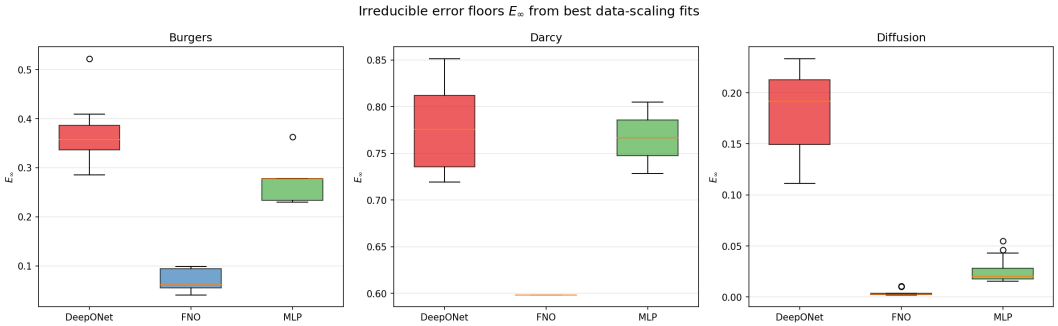

Figure 5: Estimated irreducible error floors E_∞ from data-scaling fits. FNO achieves the lowest estimated floor on every task. Darcy has uniformly high floors, indicating an intrinsically hard approximation problem. These are extrapolated estimates conditional on the winning functional form.

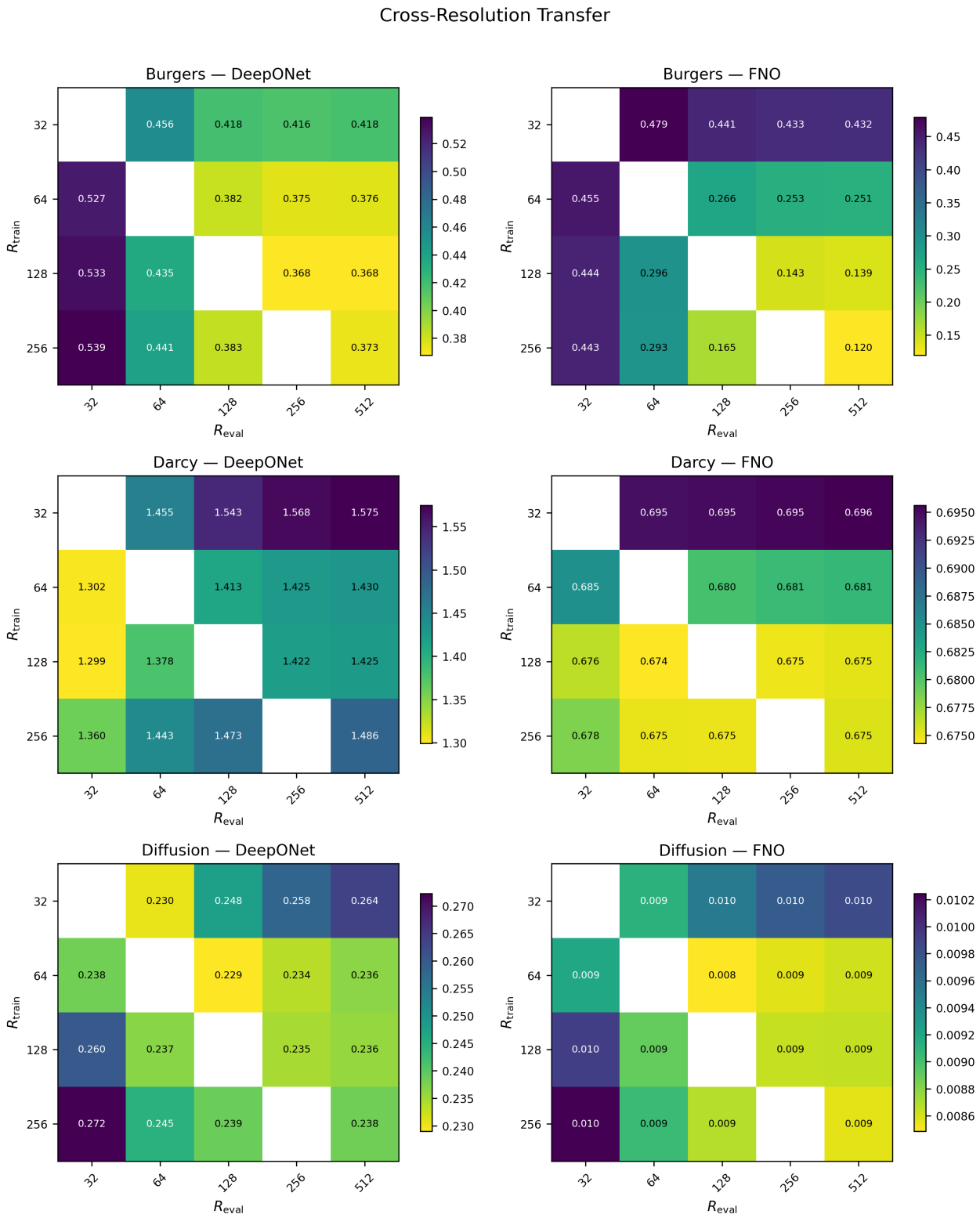

Figure 6: Cross-resolution transfer matrices. Each cell shows mean ε_{rel} for training at R_{train} (row) and evaluating at R_{eval} (column). Burgers–FNO shows strong asymmetric transfer (lower-right is best). Darcy and Diffusion are near-uniform.

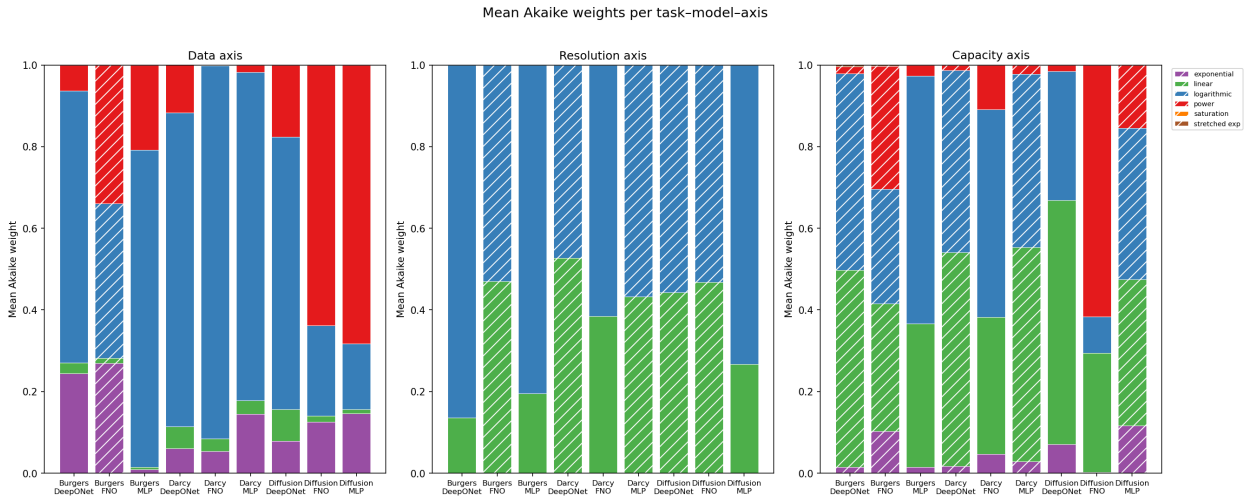

Figure 7: Mean Akaike weights across slices. Hatched bars indicate ambiguous cases where the top-two laws are within 15% of each other.

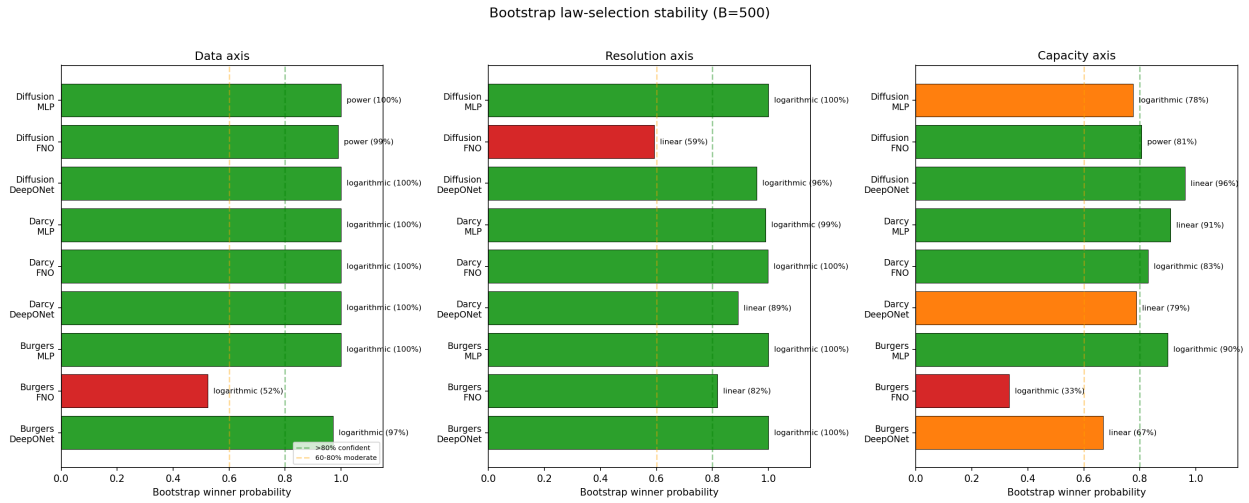

Figure 8: Bootstrap law-selection stability ($B = 500$). Green bars (>80%) indicate confident selection; orange (60–80%) moderate; red (<60%) ambiguous.